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1 Step 4: Scientific Notes and Implementation Details

1.1 Physical Constants and Quality Controls

1.1.1 Constants (SI units)

```
G = 9.80665           # Standard gravity (m s-2)
OMEGA = 7.292e-5      # Earth angular velocity (rad s-1)
R_d = 287.0           # Gas constant dry air (J kg-1 K-1)
C_p = 1004.0          # Specific heat constant pressure (J kg-1 K-1)
KAPPA = R_d / C_p    0.286 # Poisson constant
P_0 = 100000.0        # Reference pressure 1000 hPa (Pa)
R_EARTH = 6.371e6     # Earth radius (m)
```

1.1.2 Quality Control Thresholds

```
MIN_LAT = 5.0         # Minimum |lat| for EGR (avoid f→0 at equator)
MAX_EGR_DAY = 5.0     # Maximum reasonable EGR (cap unphysical values)
MIN_N_SQUARED = 1e-6  # Minimum N2 for stable stratification (s-2)
```

1.2 Eady Growth Rate (EGR) - Detailed Derivation

1.2.1 1. Virtual Temperature Correction

Standard temperature T does not account for moisture effects on air density. Virtual temperature T_v is the temperature dry air would need to have the same density as moist air at the same pressure:

$$T_v = T(1 + 0.61q)$$

where: - T = temperature (K) - q = specific humidity (kg/kg) - 0.61 comes from the ratio of gas constants: $(R_v/R_d - 1) \approx 0.608$

Physical meaning: Moisture makes air less dense (water vapor lighter than dry air), so virtual temperature is higher than actual temperature in moist air.

Typical magnitude: For subtropical cyclones with $q \sim 0.01$ kg/kg, correction is ~ 6 K.

1.2.2 2. Virtual Potential Temperature

Potential temperature θ is the temperature an air parcel would have if brought adiabatically to reference pressure $p_0 = 1000$ hPa:

$$\theta = T \left(\frac{p_0}{p} \right)^\kappa$$

Virtual potential temperature combines both corrections:

$$\theta_v = T_v \left(\frac{p_0}{p} \right)^\kappa$$

Physical meaning: θ_v is conserved for adiabatic, reversible processes in moist air. It's the natural vertical coordinate for atmospheric stability.

1.2.3 3. Brunt-Väisälä Frequency (Static Stability)

The Brunt-Väisälä frequency N characterizes the frequency of vertical oscillations of a displaced air parcel in a stably stratified atmosphere:

$$N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}$$

Derivation (simplified): 1. Vertically displace air parcel by z in environment with stratification $\partial \theta_v / \partial z$ 2. Parcel conserves its θ_v , environment has different θ_v 3. Buoyancy force: $F = -g \frac{\delta \theta_v}{\theta_v}$ 4. Equation of motion: $\frac{d^2(\delta z)}{dt^2} = -N^2(\delta z)$ 5. Solution: oscillation with frequency N (if $N^2 > 0$)

Physical interpretation: - $N^2 > 0$: Stable stratification $\rightarrow$ parcel oscillates (gravity waves) - $N^2 = 0$: Neutral stratification $\rightarrow$ parcel remains displaced - $N^2 < 0$: Unstable stratification $\rightarrow$ parcel accelerates away (convection)

Typical values: - Troposphere: $N \sim 0.01 \text{ s}^{-1}$ (period ~ 10 min) - Stratosphere: $N \sim 0.02 \text{ s}^{-1}$ (period ~ 5 min)
- Our results: $N \sim 0.0105 \pm 0.0020 \text{ s}^{-1}$

Computational implementation:

```
# 3-level centered finite difference
dtheta_v = theta_v[upper] - theta_v[lower]
dz = z[upper] - z[lower]
dtheta_v_dz = dtheta_v / dz
```

```
#  $N^2 = (g/v)(v/z)$ 
```

```

N_squared = (G / theta_v[middle]) * dtheta_v_dz
N = sqrt(N_squared) # only if N^2 > MIN_N_SQUARED

```

1.2.4 4. Vertical Wind Shear

Magnitude of the vertical wind shear vector:

$$\left| \frac{\partial \vec{V}}{\partial z} \right| = \sqrt{\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2}$$

Physical meaning: Rate of change of horizontal wind with height. Strong shear tilts developing baroclinic waves, allowing efficient extraction of potential energy.

Thermal wind balance: In geostrophic balance, vertical shear is proportional to horizontal temperature gradient:

$$\frac{\partial \vec{V}_g}{\partial z} = -\frac{g}{fT} \hat{k} \times \nabla_p T$$

This links baroclinicity (temperature gradient) to vertical shear $\rightarrow$ both appear in EGR formula.

Typical values: - Weak shear: $< 0.003 \text{ s}^{-1}$ - Moderate: $0.003\text{-}0.006 \text{ s}^{-1}$ - Strong: $> 0.006 \text{ s}^{-1}$ - Our results: $0.0051 \pm 0.0015 \text{ s}^{-1}$ (moderate to strong)

1.2.5 5. Eady Growth Rate Formula

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

Derivation: From Eady (1949) model of baroclinic instability on an f-plane with constant N and linear shear. The factor 0.31 comes from solving the eigenvalue problem for the most unstable wave.

Dimensional analysis: - $[f] = \text{s}^{-1}$ (frequency) - $[N] = \text{s}^{-1}$ (frequency) - $[\partial V / \partial z] = \text{s}^{-1}$ (shear) - $[\sigma] = \frac{\text{s}^{-1}}{\text{s}^{-1}} \cdot \text{s}^{-1} = \text{s}^{-1}$ (growth rate)

Physical interpretation:

1. $|f|$ (**rotation**): Provides restoring force for geostrophic balance. Higher latitude $\rightarrow$ stronger rotation $\rightarrow$ faster growth.
2. $1/N$ (**weak stability**): Weaker static stability allows easier vertical displacement $\rightarrow$ faster growth. Factor $1/N$ means growth rate inversely proportional to stability.
3. $|\partial V / \partial z|$ (**shear**): Tilts the wave, allowing efficient conversion of available potential energy to kinetic energy.

Why 0.31? From Eady model: - Most unstable wavelength: $\lambda = 3.9 \frac{NH}{f}$ (H = scale height) - Maximum growth rate: $\sigma_{max} = 0.31 \frac{f}{N} |\partial V / \partial z|$

Conversion to day⁻¹:

$$\sigma_{day} = \sigma_{s^{-1}} \times 86400 \text{ s/day}$$

Interpretation of values: - 0.5-1.0 day⁻¹: Moderate baroclinic instability - 1.0-2.0 day⁻¹: Strong instability (typical cyclogenesis) - 2.0-5.0 day⁻¹: Very strong instability (explosive cyclogenesis) - $> 5.0 \text{ day}^{-1}$: Likely unphysical (numerical issues or extreme events)

Our results: - Mean: 1.17-1.35 day⁻¹ (strong baroclinic instability) - Max: 2.44-4.11 day⁻¹ (very strong in most intense regions)

1.3 Rayleigh-Kuo (RK) Criterion - Detailed Derivation

1.3.1 1. Relative Vorticity

Vertical component of vorticity (rotation rate about vertical axis):

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Spherical coordinates (Earth is not flat!):

$$\frac{\partial}{\partial x} = \frac{1}{R_E \cos \phi} \frac{\partial}{\partial \lambda}$$
$$\frac{\partial}{\partial y} = \frac{1}{R_E} \frac{\partial}{\partial \phi}$$

where ϕ = latitude, λ = longitude.

Physical meaning: Measures local spinning of air. Positive for counterclockwise rotation (cyclonic in NH, anticyclonic in SH).

Typical values: - Synoptic systems: $|\zeta| \sim 10^{-5} \text{ s}^{-1}$ - Strong cyclones: $|\zeta| \sim 10^{-4} \text{ s}^{-1}$

1.3.2 2. Absolute Vorticity

Sum of relative vorticity and planetary vorticity:

$$\eta = \zeta + f$$

where $f = 2\Omega \sin \phi$ is the Coriolis parameter.

Physical meaning: Total rotation rate as seen from inertial (non-rotating) frame. Combines Earth's rotation (f) with atmospheric rotation (ζ).

Why “absolute”? It's the vorticity in an absolute (inertial) reference frame, not rotating with Earth.

At 45°S: $f \approx -10^{-4} \text{ s}^{-1}$ (negative in SH)

1.3.3 3. Rayleigh-Kuo Criterion

Statement: A necessary condition for barotropic instability is that the meridional gradient of absolute vorticity **changes sign** somewhere in the domain:

$$\frac{\partial \eta}{\partial y} = 0 \text{ somewhere, with } \frac{\partial \eta}{\partial y} < 0 \text{ on one side and } > 0 \text{ on other}$$

Or more simply: $\min(\partial \eta / \partial y) < 0$ AND $\max(\partial \eta / \partial y) > 0$

Historical context: - **Rayleigh (1880):** For inviscid parallel shear flow, instability requires $\frac{d^2 U}{dy^2}$ to change sign - **Kuo (1949):** Extended to rotating fluids: criterion becomes sign change of $\frac{\partial \eta}{\partial y}$

Physical mechanism:

1. **Basic state:** Flow with absolute vorticity gradient $\partial \eta / \partial y$
2. **Perturbation:** Small wave-like disturbance
3. **Vorticity equation:** Perturbation vorticity evolves as $\frac{\partial \zeta'}{\partial t} + V \frac{\partial \zeta'}{\partial y} = -v' \frac{\partial \eta}{\partial y}$
4. **Growth:** If $\partial \eta / \partial y$ changes sign, perturbations can extract energy from basic flow

Why sign change? Counter-propagating Rossby waves on the two sides of the $\partial \eta / \partial y = 0$ point can phase-lock and amplify each other $\rightarrow$ instability.

Interpretation: - Criterion is **necessary** but **not sufficient** - Satisfaction means instability is *possible* (if other conditions met) - Non-satisfaction means instability is *impossible*

1.3.4 4. Connection to Barotropic Energy Conversion

Barotropic conversion C_k represents kinetic energy transfer from eddies to mean flow (or vice versa):

$$C_k = -\overline{u'v'} \frac{\partial \bar{u}}{\partial y} - \overline{v'v'} \frac{\partial \bar{v}}{\partial y}$$

Physical link to RK: - RK criterion indicates regions where wave growth can occur via horizontal shear - Growing waves generate Reynolds stresses $\overline{u'v'}$ - These stresses cause momentum transport and energy conversion - Negative C_k (common in cyclones) means energy flows from mean to eddies

Our finding: All 10 EP1 cyclones satisfy RK criterion at all scales (5°, 15°, 30°). **Implication:** Favorable conditions for barotropic processes are widespread, not just locally confined.

1.4 Results Interpretation

1.4.1 Eady Growth Rate Statistics

Domain	Size	Mean EGR (day ⁻¹)	Std Dev	Max EGR (day ⁻¹)
Local	5°	1.237	0.441	2.442 ± 0.668
Mesoscale	15°	1.353	0.295	3.518 ± 0.673
Synoptic	30°	1.169	0.178	4.111 ± 0.643

Key findings: 1. **Strong baroclinic instability:** EGR $\sim$ 1.2-1.4 day⁻¹ indicates rapid growth potential 2. **Scale consistency:** EGR values similar across scales $\rightarrow$ baroclinic processes are large-scale 3. **Spatial structure:** Maximum EGR increases with domain size (local max < mesoscale max < synoptic max) - Suggests strongest instability not concentrated at cyclone center - Rather, extends across broader synoptic environment 4. **Variability:** Higher variance at local scale ($\sigma = 0.44$) than synoptic ($\sigma = 0.18$) - Local conditions more variable case-to-case - Large-scale baroclinicity more consistent

Comparison with literature: - Gyakum et al. (1989): EGR $\sim$ 1-2 day⁻¹ for explosive cyclogenesis - Our values (1.2-1.4 day⁻¹) consistent with intensifying extratropical cyclones

1.4.2 Rayleigh-Kuo Criterion Results

Domain	Size	Cases Satisfied	Percentage
Local	5°	10/10	100%
Mesoscale	15°	10/10	100%
Synoptic	30°	10/10	100%

Key findings: 1. **Universal satisfaction:** All EP1 cyclones meet necessary condition for barotropic instability 2. **Scale independence:** Criterion satisfied at all scales examined 3. **Contrary to hypothesis:** Initial expectation was local-scale phenomenon - Results show large-scale flow configuration supports barotropic processes - Suggests interaction between synoptic forcing and local cyclone dynamics

Physical interpretation: - Large-scale flow pattern (jets, fronts) creates $\partial\eta/\partial y$ structure - This structure persists across scales from cyclone core to synoptic environment - Both local (cyclone-induced) and remote (synoptic) contributions to vorticity gradient

Connection to Ck: - Mean Ck for EP1: -16.48 W/m² (strongest barotropic conversion of all EPs) - RK satisfaction indicates this conversion occurs through barotropic instability mechanism - Negative Ck means energy flows from mean flow → eddies → consistent with RK-based wave growth

1.5 Computational Considerations

1.5.1 Vertical Derivatives

Centered finite differences used for accuracy:

$$\frac{\partial f}{\partial z} \approx \frac{f_{i+1} - f_{i-1}}{z_{i+1} - z_{i-1}}$$

Requires 3 vertical levels (upper, middle, lower).

Advantages: - Second-order accurate: $O(\Delta z^2)$ - Symmetric: no bias toward upper or lower level

Pressure to height conversion:

$$z = \frac{\Phi}{g}$$

where Φ = geopotential (m² s⁻²), downloaded from ERA5.

1.5.2 Horizontal Derivatives

Spherical geometry accounted for using metric factors:

```
dx = R_EARTH * cos(lat) * dlon_radians
dy = R_EARTH * dlat_radians
```

```
du_dy = gradient(u, axis=0) / dy
dv_dx = gradient(v, axis=1) / dx
```

Why necessary? - On sphere, 1° longitude 1° latitude in metric distance - At equator: 1° lon 111 km, 1° lat 111 km - At 45°: 1° lon 78 km, 1° lat 111 km

Failure to account for this introduces systematic errors in vorticity calculation.

1.5.3 Quality Control Flags

1. **Near-equator masking:** $|lat| < 5^\circ$ excluded
 - $f \rightarrow 0$ near equator makes f/N term unstable
 2. **Static stability check:** Require $N^2 > 10^{-6} \text{ s}^{-2}$
 - Filters unstable stratification (convection, not baroclinic instability)
 - Also filters numerical noise in $\partial\theta_v/\partial z$
 3. **EGR cap:** Values $> 5.0 \text{ day}^{-1}$ set to NaN
 - Removes unphysical extremes from averaging
 - Can occur near topography, data gaps, or numerical issues
-

1.6 References

Eady Model: - Eady, E. T. (1949). Long waves and cyclone waves. *Tellus*, 1(3), 33-52.

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